‘Yes, the Copenhagen interpretation is weird, but so is everything else.’ And you sort of want to slap them and say, ‘No! The Copenhagen interpretation is not *weird*, it’s gibberish, it’s unintelligible.’”

And some physicists argue, like good positivists, that because no experiment can tell the difference between the different interpretations, it’s meaningless to draw any distinctions between them—that even if Copenhagen is inconsistent, it doesn’t matter which, if any, alternative to it we adopt. And this is simply not true. If we want to go beyond our current theories to devise a new theory, to find new physics and explain new experimental results, our interpretations matter. Ask two physicists, a pilot-wave theorist and a many-worlds theorist, what kind of theory they expect to see going beyond quantum physics, and you’ll get two very different answers. Richard Feynman pointed out that although there’s no experimental way to tell the difference between two mathematically equivalent theories (i.e., two different interpretations of the same math), subscribing to one theory or the other makes a huge difference in how you think about the world. That difference, in turn, affects the new ideas and new theories we develop. For example, the sixteenth-century astronomer Tycho Brahe had a theory in which the Earth was at the center of the universe, the Sun and Moon orbited the Earth, and the rest of the planets orbited the Sun. His theory was mathematically equivalent to the heliocentric model of Copernicus—it gave identical predictions about the movements of lights in the sky—but the idea that the Earth wasn’t at the center of the universe led to an entirely different set of ideas about how the universe works. Similarly, we could develop an interpretation of quantum physics in which wave functions are powered by invisible unicorns, which obey aggregate laws of herding and flocking that lead to the Schrödinger equation. But (I hope) we can agree that this is a bad idea, significantly worse than any other interpretation. Experimental results are not the only things that enter into the formulation and evaluation of scientific theories, nor could they be. The full content of our theories—not only the mathematics but the claims about the nature of the world that come along with the mathematics—is important to the work of science.

And the worldviews we get from our best scientific theories also make their way to the public and inform how we see ourselves, as I argued in the Introduction. This has already happened with the Copenhagen interpretation